

ZYDUS LIFESCIENCES LTD.

Executive Architecture Whitepaper & GxP Audit Dossier



Predictive Maintenance & Oncology Asset Intelligence Platform

Ref: ZYDUS-ENG-WP-2026-V3 | Standard: US FDA 21 CFR Part 11 / EU Annex 11 / GAMP 5 Cat 4

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1. EXECUTIVE SUMMARY & MISSION-CRITICAL OBJECTIVES

ZyduS Lifesciences operates high-throughput pharmaceutical oral solid dosage facilities, sterile injectable complexes, biologics cold-chain storage, and advanced oncology radiation therapy centers. Unplanned equipment failure exposes production batches valued between **INR 12,50,000 and INR 9,50,00,000** to catastrophic loss and risks patient treatment interruptions. This platform implements autonomous industrial telemetry ingestion, physics-informed machine learning, and a mathematically immutable **US FDA 21 CFR Part 11 compliant SHA-256 audit trail**.

2. REGULATORY COMPLIANCE MATRIX

Standard	Mandate & Section	Implementation Mechanism
US FDA 21 CFR Part 11	Section 11.10(e) Audit Trails	SHA-256 Immutable Hash Chaining ($H_i = \text{SHA-256}(H_{i-1} \text{payload})$).
US FDA 21 CFR Part 11	Section 11.50 Electronic Signatures	Dual-factor authentication modal with password re-entry & legal perjury certification.
ISPE GAMP 5	Category 4 Software	84 Pytest unit/integration tests (100% green), Playwright E2E browser automation.
EU GMP Annex 11	Section 9 Data Integrity	Automated continuous cryptographic verification via <code>/api/audit-logs/verify</code> .

3. MATHEMATICAL & MACHINE LEARNING FOUNDATIONS

- Digital Twin Health Index (DTHI):** $DTHI(t) = 100 * (1.0 - [0.50 P_{fail} + 0.30 S_{anomaly} + 0.20 D_{physics}])$
- GAMP 5 Batch Exposure Risk (INR):** $Loss\ Exposure = Batch\ Value\ (INR) * P_{failure}(t)$
- Explainable AI (SHAP):** Computes Shapley values to identify root-cause sensor drivers with percentage attribution.
- Population Stability Index (PSI):** Quantifies distribution shift across sensor channels with a $PSI \geq 0.25$ autonomous retraining trigger.

4. 20-ASSET DIGITAL TWIN FLEET REGISTRY (EXECUTIVE SUMMARY)

Asset Tag	Equipment Name	Facility Block	Batch Value (INR)	Key Sensors
GRAN-LINE-01	High Shear Mixer Granulator 600L	Oral Solid Block A	INR 25,00,000	Vibration, Temp, Current, RPM
TABLET-PRESS-01	High Speed Rotary Tablet Press	Oral Solid Block A	INR 18,50,000	Compression, Displacement, Feeder
ASEPTIC-FILL-01	Aseptic Isolator Liquid Vial Filler	Sterile Complex B	INR 65,00,000	Diff Pressure, Fill Acc, Temp
ULT-FREEZER-01	Ultra-Low Temp Biobank (-86C)	Biologics Plant C	INR 85,00,000	Chamber Temp, Power, Doors
COLD-ROOM-01	Vaccine Cold Storage (2-8C)	Biologics Plant C	INR 1,20,00,000	Ambient Temp, RH%, Defrost
HPLC-STACK-01	Quaternary UPLC Chromatography	Central QC Lab	INR 15,00,000	Pressure, Flow Rate, Oven Temp
LCMS-01	Triple Quadrupole LC-MS/MS	Central QC Lab	INR 90,00,000	Source Temp, Gas Flow, Vacuum
LINAC-01	Medical Linear Accelerator (6-18 MeV)	Cancer Center	INR 9,50,00,000	Beam Current, Dose Rate, Arc V
CT-SCANNER-01	128-Slice Oncology CT Simulator	Cancer Center	INR 6,00,00,000	Anode Temp, Rotor RPM, kV

5. GxP VALIDATION & REGULATORY SIGN-OFF

Role	Signer Name & Title	Date	21 CFR Status
Lead AI/ML Architect	Vedant Panchal, Principal AI Engineer	01-Sep-2026	CERTIFIED_ACTIVE
GxP Quality Auditor	Dr. A. Sharma, VP Quality & Compliance	01-Sep-2026	21_CFR_PART_11_SIGNED
Plant Operations Director	R. Patel, Head of Global Oncology Mfg	01-Sep-2026	21_CFR_PART_11_SIGNED

CERTIFICATE OF IMMUTABILITY: Generated and cryptographically validated by ZyduS Predictive Maintenance Platform v3.0.0. Audit Ledger Status: SECURE_IMMUTABLE (Zero Mathematical Deviations).